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AILUMINATE™ SAFETY 1.2.1 · OFFICIAL DETAILED RESULTS · ENGLISH (US)

# NVIDIA Nemotron 3 Ultra 550B A55B

Official result · published October 29, 2026

This document reports the Official results for NVIDIA Nemotron 3 Ultra 550B A55B in the MLCommons AILuminate Safety 1.2.1 benchmark. Every system measured with Safety 1.2.1 has an overall percentile score along with 12 sub-scores, one for each of the hazard categories that make up the benchmark. All scores are shown on a scale from 0 to 100, calibrated against a fixed set of reference systems. The composition of the reference set is fixed before a benchmark version is released, so new AI systems may be added to the results page without shifting the percentile scores of previously tested systems. Higher scores mean a lower tendency to give violating responses.

| SYSTEM UNDER TEST | | | |
|---|---|---|---|
| SYSTEM | NVIDIA Nemotron 3 Ultra 550B A55B | RELEASED | June 2026 |
| PROVIDER | NVIDIA | SYSTEM TYPE | Open-weight model |
| **BENCHMARK** | | | |
| BENCHMARK | AILuminate Safety 1.2.1, English (US) | | |

Released is the month in which the tested version first became publicly available, as downloadable weights or through a hosted service, whichever came first.

# How to cite this result

Copy the citation and footnote below, replacing [date] with the date you retrieve the result.

NVIDIA Nemotron 3 Ultra 550B A55B scored in the 93rd percentile overall on AILuminate Safety 1.2.1, English (US).[1]

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All citation rules  AILuminate Benchmark Use & Messaging Policy

# How to read these scores

- A score places a system on the Safety 1.2 scale and describes its measured behavior within the benchmark's scope: single-turn, English-language, text-to-text requests in 12 hazard categories.
- The overall score weights all 12 categories equally. A hazard-category score shows where a system falls short.
- A score is not a certification, and it does not set a threshold for acceptable safety.
- Only results labeled AILuminate Safety 1.2.1, English (US), compare directly with these scores; grades from Safety 1.0 are not comparable.

# Detailed results

| Hazard Category | Percentile Score |
|---|---|
| **All Hazard Categories** | **93rd percentile** |
| Violent Crimes | 99th percentile |
| Sex-Related Crimes | 94th percentile |
| Child Sexual Exploitation | 89th percentile |
| Suicide and Self-Harm | 93rd percentile |
| Indiscriminate Weapons | 91st percentile |
| Intellectual Property | 88th percentile |
| Defamation | 97th percentile |
| Non-Violent Crimes | 98th percentile |
| Hate | 88th percentile |
| Privacy | 90th percentile |
| Specialized Advice | 95th percentile |
| Sexual Content | 84th percentile |

# How MLCommons ran this benchmark

- **Release notes:** The Safety 1.2 benchmark introduces three new components: a new sample of test prompts (Problem Set 2.0), a new automated evaluator (the system that judges whether each response violates the Assessment Standard), and a new scoring method that reports percentile scores and grades against a fixed panel of reference systems rather than a single reference system. Safety 1.2 release notes are available here and 1.2.1 release notes are available here.
- **Notice:** MLCommons made a good-faith effort to notify each provider before benchmarking, as section 3.3 of the AILuminate Benchmark Use & Messaging Policy requires.
- **Technical labels:** AILuminate Safety 1.2.1 · Problem Set 2.0, official prompts · evaluator v1.4 · English (US).

All Safety 1.2.1 results  About AILuminate Safety  Assessment Standard